In [1]:
```python
import pandas as pd
```

In [3]:
```python
#Task 1 Load IPL data

ipl = pd.read_csv("IPL_2023_Batsmen.csv")

print(ipl.head(10))
```

```
         player_name  runs  matches  average  strike_rate
0       Shubman Gill   890       17    59.33       157.80
1     Faf du Plessis   730       14    56.15       153.68
2       Devon Conway   672       16    51.69       139.70
3        Virat Kohli   639       14    53.25       139.82
4  Yashasvi Jaiswal   625       14    48.08       163.61
5    Ruturaj Gaikwad   590       16    42.14       147.50
6       David Warner   516       14    36.85       130.12
7        Rinku Singh   474       14    59.25       149.05
8  Suryakumar Yadav   605       16    43.21       181.32
9        Jos Buttler   392       14    28.00       139.50
```

In [4]:
```python
# TASK 2 - Univariate Analysis

print("Mean:", ipl["runs"].mean())
print("Median:", ipl["runs"].median())
print("Mode:", ipl["runs"].mode()[0])
print("Minimum:", ipl["runs"].min())
print("Maximum:", ipl["runs"].max())
```

```
Mean: 513.875
Median: 495.0
Mode: 392
Minimum: 210
Maximum: 890
```

In [5]:
```python
# TASK 3 Zomato Correlation

zomato = pd.DataFrame({"name": ["Dominos", "KFC", "McDonalds", "Subway", "Pizza
"votes": [1200, 900, 1500, 800, 700],"avg_cost": [500, 450, 400, 350, 550]})

print(zomato["rating"].corr(zomato["votes"]))
```

```
0.9183996645933807
```

In [8]:
```python
# TASK 4 Summary Statistics

print(ipl.describe())
```

```
             runs    matches    average  strike_rate
count   16.000000  16.000000  16.000000    16.000000
mean   513.875000  14.062500  41.736875   150.762500
std    176.840371   2.235136  12.086389    14.251558
min    210.000000   8.000000  20.750000   130.120000
25%    383.500000  14.000000  30.240000   139.790000
50%    495.000000  14.000000  42.505000   148.275000
75%    628.500000  16.000000  52.080000   159.252500
max    890.000000  17.000000  59.330000   181.320000
```

In [13]:
```python
print("Insight: The average runs scored by the players are 513.88, while the hig
```

```
Insight: The average runs scored by the players are 513.88, while the highest run
s are 890. This shows that some players scored significantly more runs than the a
verage..
```

In [12]:

```python
# TASK 5

# Two suggested analyses:
# 1. Correlation between runs and strike rate
# 2. Correlation between runs and matches

print("Correlation between runs and strike rate:")
print(ipl["runs"].corr(ipl["strike_rate"]))
```

```
Correlation between runs and strike rate:
0.1329808103764836
```

#Analysis I performed a bivariate analysis between runs and strike rate. The correlation coefficient shows the relationship between a player's total runs and strike rate. “A positive correlation means that players with higher strike rates tend to score more runs.”